

PINE RIVER, MN, USA

WMO: 727508

Lat: 46.725N

Lon: 94.382W

Elev: 1295

StdP: 14.02

Time Zone: -6.00 (NAC)

Period: 99-19

WBAN: 54915

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-22.0	-16.7	-27.1	1.3	-21.6	-21.7	1.7	-14.1	18.8	14.5	17.3	15.5	1.2	300	0.574

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	21.9	87.6	72.7	83.6	70.5	81.2	68.5	75.5	83.5	73.3	80.4	71.1	77.8	9.6	170

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
72.8	127.8	80.0	71.7	122.7	79.1	69.6	114.0	76.7	40.0	81.9	37.9	80.3	35.7	78.4	82.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 18.6	16.7	15.0	DB	-29.6	91.1	6.1	3.5	-34.0	93.6	-37.6	95.6	-41.0	97.5	-45.4	100.0	(4)
(5)			WB	-30.3	78.1	6.0	2.7	-34.7	80.0	-38.2	81.6	-41.6	83.1	-46.0	85.1	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	40.9	10.2	13.0	27.6	41.4	53.6	63.0	68.3	64.8	57.0	43.6	30.6	15.2		
	DBStd	22.60	14.01	12.99	12.84	9.83	8.21	6.45	5.89	5.90	8.50	8.79	10.68	12.94		
	HDD50	5239	1234	1036	698	287	54	1	0	0	27	239	585	1078		
	HDD65	9111	1699	1456	1159	708	365	110	33	76	265	664	1033	1543		
	CDD50	1901	0	0	4	30	166	392	569	460	237	41	2	0		
	CDD65	298	0	0	0	1	12	51	137	71	25	1	0	0		
	CDH74	3139	0	0	0	18	198	536	1421	708	237	21	0	0		
	CDH80	794	0	0	0	3	47	113	413	166	49	3	0	0		
	WSAvg	5.9	5.5	5.7	6.3	7.3	7.0	5.9	4.9	4.7	5.6	6.0	6.0	5.6		
	PrecAvg	28.0	0.7	0.6	1.3	2.1	3.2	4.3	4.1	3.1	3.1	3.1	1.3	0.9		
(15) Precipitation	PrecMax	33.7	1.9	1.9	3.5	5.2	7.6	7.5	7.6	6.3	7.1	6.0	4.1	2.1		
	PrecMin	20.8	0.1	0.0	0.4	0.8	1.2	1.5	1.9	0.9	0.7	0.4	0.2	0.3		
	PrecStd	3.3	0.5	0.4	0.9	1.0	1.5	1.6	1.6	1.5	1.3	1.6	0.9	0.5		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	44.7	46.4	65.6	75.4	85.5	88.0	91.2	90.2	85.6	76.9	63.4	43.5		
		MCWB	38.7	40.9	56.7	56.7	65.9	73.7	75.3	74.2	70.2	61.4	53.8	38.8		
	2%	DB	36.5	41.1	56.9	69.9	79.5	82.4	88.1	84.0	80.9	69.7	55.4	38.6		
		MCWB	33.4	36.0	47.9	53.3	61.5	69.5	73.7	71.0	68.0	58.5	48.2	35.9		
	5%	DB	33.7	36.4	50.4	64.1	74.6	80.7	84.0	81.3	75.3	63.4	51.7	35.8		
		MCWB	32.0	33.4	44.4	49.9	59.2	67.1	71.7	68.9	65.2	54.4	44.6	33.9		
	10%	DB	29.7	32.3	45.3	59.5	70.3	76.7	81.6	78.7	72.3	57.4	45.8	32.5		
		MCWB	28.9	30.6	40.4	46.6	57.3	64.4	69.8	67.2	63.8	50.7	41.6	31.6		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	38.7	42.2	57.3	60.3	70.3	76.0	79.2	77.8	73.9	64.2	55.3	39.6		
		MCDB	43.9	46.7	60.9	73.4	80.4	84.0	85.7	85.7	81.3	70.1	59.7	41.9		
	2%	WB	34.1	37.1	49.6	55.2	65.7	73.0	76.4	73.7	70.6	59.7	50.0	36.3		
		MCDB	36.2	40.4	56.5	65.5	74.5	80.4	84.5	80.7	76.9	67.7	56.4	38.6		
	5%	WB	31.5	33.7	44.8	51.7	62.4	69.8	74.4	71.3	67.8	55.5	45.6	34.0		
		MCDB	33.0	36.4	49.8	62.4	70.5	76.6	81.9	77.9	73.4	61.8	49.0	35.2		
	10%	WB	28.3	30.3	40.2	48.3	59.2	66.9	72.0	69.1	65.2	52.3	41.7	31.4		
		MCDB	28.9	32.1	45.3	58.3	67.1	73.8	78.7	75.5	70.5	57.3	46.6	32.6		
(35) Mean Daily Temperature Range	5% DB	MDBR	19.3	20.7	20.0	21.3	22.7	21.0	21.9	22.0	21.5	18.4	16.6	15.8		
		MCDBR	19.1	21.8	26.4	31.2	31.1	25.1	25.5	25.3	26.5	27.6	25.6	18.9		
		MCWBR	16.3	17.1	17.5	17.5	16.3	13.3	13.2	13.6	15.0	16.9	17.1	15.8		
		MCDBR	18.4	20.0	25.4	27.6	24.4	20.8	22.7	21.6	20.8	23.2	22.1	17.0		
	5% WB	MCWBR	15.9	15.9	17.6	16.9	14.8	12.8	13.4	13.1	13.3	15.4	16.0	15.0		
(40) Clear-Sky Solar Irradiance	taub		0.256	0.288	0.308	0.356	0.374	0.375	0.379	0.385	0.354	0.311	0.283	0.263		
	taud		2.374	2.285	2.386	2.348	2.354	2.382	2.388	2.387	2.453	2.564	2.517	2.398		
	Ebn at Noon		270	280	292	285	283	282	279	273	273	272	260	253		
	Edh at Noon		22	30	32	37	38	37	36	35	30	23	20	19		
(44) All-Sky Solar Radiation	RadAvg		504	862	1215	1529	1700	1857	1976	1662	1239	762	468	351		
	RadStd		56	87	121	131	121	126	86	139	93	116	48	43		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	-4.54	N/A	-2.03	-2.27	-2.20	N/A	N/A	N/A	N/A
Regional (46 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page